

# SoS 2026: Algorithmic Trading

## Plan of Action

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### Timeline

- Week 4**    Electronic Markets & the Limit Order Book: how electronic markets function, order types, exchange structures, and the mechanics of the limit order book. (*Cartea Ch. 1*)
- Week 5**    Market Microstructure: market making models, trading costs, liquidity measurement, adverse selection, and price formation. (*Cartea Ch. 2*)
- Week 6**    Stochastic Optimal Control: the mathematical framework underlying algorithmic execution strategies: dynamic programming, Hamilton, Jacobi Bellman equations, and optimal stopping. (*Cartea Ch. 5*)
- Week 7**    Market Making: inventory-based and adverse-selection market-making models; how market makers choose quotes and manage risk. (*Cartea Ch. 10*)

### References

*Algorithmic and High-Frequency Trading* by Álvaro Cartea, Sebastián Jaimungal & José Penalva